IBM DATA SCIENCE PROFESSIONAL CERTIFICATE · CAPSTONE

Predicting SpaceX Falcon 9 First-Stage Landings

Data collection, exploratory analysis, interactive maps and machine learning on 90 Falcon 9 launches (2010–2020)

Greg October 16, 2026

OUTLINE

What this report covers

01 Executive Summary

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03 Methodology

04 Results: EDA with visualization and SQL

05 Results: Folium maps and Plotly Dash

06 Predictive analysis, conclusion and appendix

EXECUTIVE SUMMARY

Landing success is predictable with 94% test accuracy

Methods

- Collected launch data from the SpaceX REST API and scraped Wikipedia launch tables
- Cleaned and wrangled the data; created the binary landing label **Class**
- Explored patterns with charts and SQL queries
- Built a Folium site map and a Plotly Dash dashboard
- Trained and tuned Logistic Regression, SVM, Decision Tree and KNN with GridSearchCV

Key results

- Overall landing success rate: **66.7%** (60 of 90)
- Success rose from 0% (2010–13) to **90%** in 2019
- KSC LC-39A has the highest site success rate: **77.3%**
- ES-L1, GEO, HEO and SSO orbits: **100%** success
- Best model: **Decision Tree**, 94.4% test accuracy

INTRODUCTION

Reusing the first stage is what makes Falcon 9 cheap

Background

SpaceX advertises Falcon 9 launches at about $62 million, while other providers charge upward of $165 million. Most of the saving comes from landing and reusing the first stage.

If we can predict whether the first stage will land, we can estimate the cost of a launch. A competing company, Space Y, could use that to bid against SpaceX.

Problems to answer

1. Which factors (payload, orbit, launch site, booster, flight count) affect landing success?
2. How has landing success changed over time?
3. Can a machine learning model predict whether the first stage will land?

01

Methodology

From raw API responses to a tuned classifier

METHODOLOGY

Executive summary of the methodology

Step	What was done
Data collection	SpaceX REST API (v4) requests plus BeautifulSoup scraping of Wikipedia's Falcon 9 launch tables
Data wrangling	Filtered to Falcon 9, imputed missing payload mass with the mean, built the Class label from landing outcomes
EDA	Scatter, bar and line charts with Seaborn/Matplotlib; SQL queries in SQLite
Interactive analytics	Folium map of sites, outcomes and proximities; Plotly Dash dashboard
Predictive analysis	One-hot encoding, standardization, 80/20 split, GridSearchCV (10-fold) on four classifiers

GitHub repository: github.com/gcastor/spacex-falcon9-capstone

DATA COLLECTION – SPACEX API

Launch records pulled from the SpaceX REST API

1
GET past launches
requests.get() on the v4 launches endpoint

2
Parse JSON
.json() and json_normalize into a DataFrame

3
Enrich IDs
Extra calls for rocket, launchpad, payload and core details

4
Filter Falcon 9
Drop Falcon 1 rows, reset flight numbers

5
Clean and save
Mean-impute PayloadMass, export CSV

```
spacex_url = "https://api.spacexdata.com/v4/launches/past"
response = requests.get(spacex_url)
data = pd.json_normalize(response.json())
```

GitHub: github.com/gcastor/spacex-falcon9-capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

DATA COLLECTION – WEB SCRAPING

Launch tables scraped from Wikipedia

1
Request page
GET the Falcon 9 launch list page

2
Parse HTML
Create a BeautifulSoup object

3
Get headers
find_all('table') and extract column names from th

4
Parse rows
Loop each tr and fill a launch dictionary

5
Build DataFrame
Convert the dictionary and export CSV

```
soup = BeautifulSoup(requests.get(static_url).text, "html.parser")
html_tables = soup.find_all("table")
df = pd.DataFrame(launch_dict)
```

GitHub: github.com/gcastor/spacex-falcon9-capstone/blob/main/jupyter-labs-webscraping.ipynb

DATA WRANGLING

Turning eight landing outcomes into one label

1. **Profile the data**: 90 rows × 17 columns; check nulls and dtypes
2. **Count values**: launches per site, orbits and outcomes with value_counts()
3. **Define bad outcomes**: False Ocean, False ASDS, False RTLS, None ASDS, None None
4. **Create Class**: 0 if outcome is bad, otherwise 1; export dataset_part_2.csv

55 · 22 · 13

Launches at CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E

27

GTO launches, the most common orbit

66.7%

Mean of Class: landing success rate

GitHub: github.com/gcastor/spacex-falcon9-capstone/blob/main/labs-jupyter-spacex-Data-wrangling.ipynb

EDA WITH DATA VISUALIZATION

Six charts, each answering one question

Chart type	Variables	Why it was used
Scatter	Flight Number vs Launch Site	Does experience at a site improve landings?
Scatter	Payload Mass vs Launch Site	Do sites handle heavy payloads differently?
Bar	Success rate by Orbit	Which orbits land most reliably?
Scatter	Flight Number vs Orbit	Does orbit success depend on experience?
Scatter	Payload Mass vs Orbit	How does payload interact with orbit?
Line	Yearly success rate	Is landing success trending upward?

GitHub: github.com/gcastor/spacex-falcon9-capstone/blob/main/edadataviz.ipynb

EDA WITH SQL

Ten SQL queries run against the launch table in SQLite

1. Unique launch site names
2. Five records at sites beginning with “CCA”
3. Total payload mass carried per launch site
4. Average payload mass of Block 1 boosters
5. Date of the first successful ground-pad landing
6. Boosters with drone-ship success, payload 4,000–6,000 kg
7. Total successful vs failed landings
8. Boosters that carried the maximum payload (subquery)
9. 2015 drone-ship failures by month
10. Landing outcomes ranked, 2010-06-04 to 2017-03-20

Techniques: SELECT DISTINCT, LIKE, SUM / AVG / MIN / COUNT, GROUP BY, ORDER BY, subqueries, date filtering with substr() and BETWEEN.

GitHub: github.com/gcastor/spacex-falcon9-capstone/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

INTERACTIVE VISUAL ANALYTICS

A Folium map and a Plotly Dash dashboard

Folium map

- **Circles and markers** at each launch site's coordinates
- **MarkerCluster** of every launch, green for success and red for failure
- **MousePosition and PolyLines** to measure distance to coastline, railway, highway and city

lab_jupyter_launch_site_location.ipynb

Plotly Dash app

- **Dropdown** to pick all sites or a single site
- **Pie chart** of successful launches by site, or success vs failure for one site
- **Range slider** on payload mass driving a payload vs outcome **scatter**, colored by booster

spacex_dash_app.py

GitHub: github.com/gcastor/spacex-falcon9-capstone (both files in the repository root)

PREDICTIVE ANALYSIS (CLASSIFICATION)

Building, tuning and evaluating four classifiers

1
Encode features
One-hot Orbit, Site, Pad, Serial: 80 features

→

2
Standardize
StandardScaler on all features

→

3
Split
80/20 train/test, random_state=2 (18 test rows)

→

4
Tune
GridSearchCV, 10-fold, for LR, SVM, Tree, KNN

→

5
Evaluate
Test accuracy and confusion matrix; pick best model

Grids searched: C and solver (LR); kernel, C and gamma (SVM); criterion, splitter, max_depth and leaf/split sizes (Tree); n_neighbors, algorithm and p (KNN).

GitHub: github.com/gcastor/spacex-falcon9-capstone/blob/main/SpaceX_Machine_Learning_Prediction_Part_5.ipynb

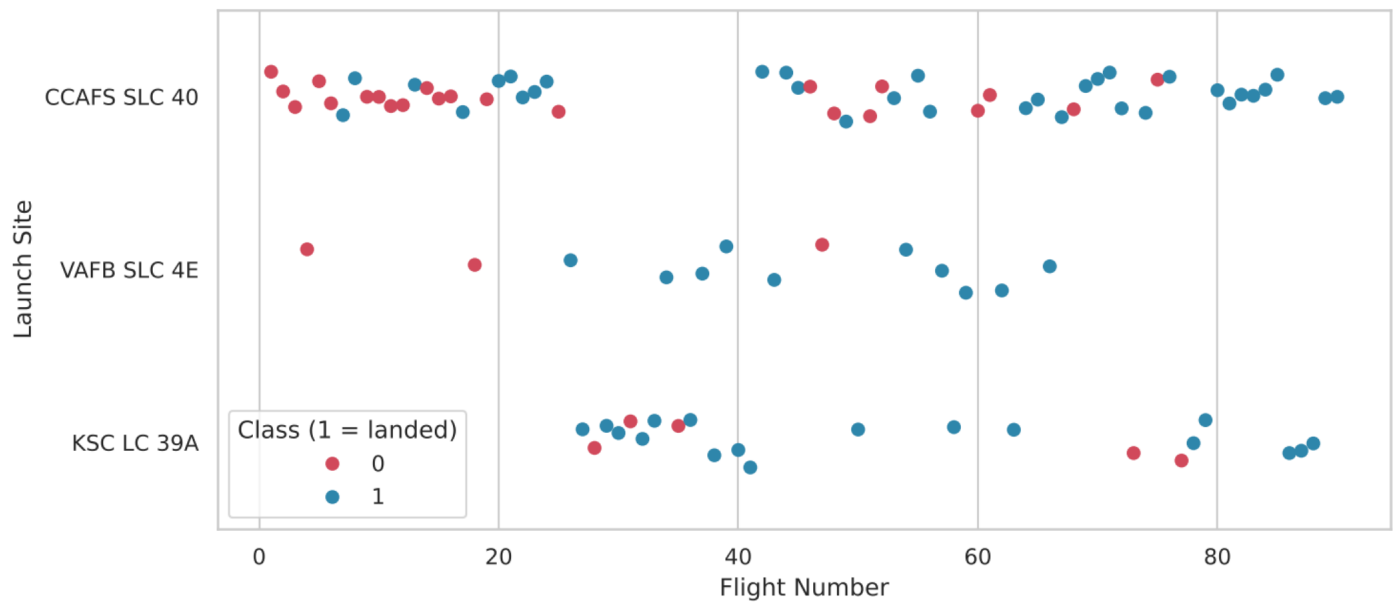
02

Insights from Exploratory Data Analysis

Visualization results, then SQL results

EDA · FLIGHT NUMBER VS LAUNCH SITE

Later flights land more often at every site



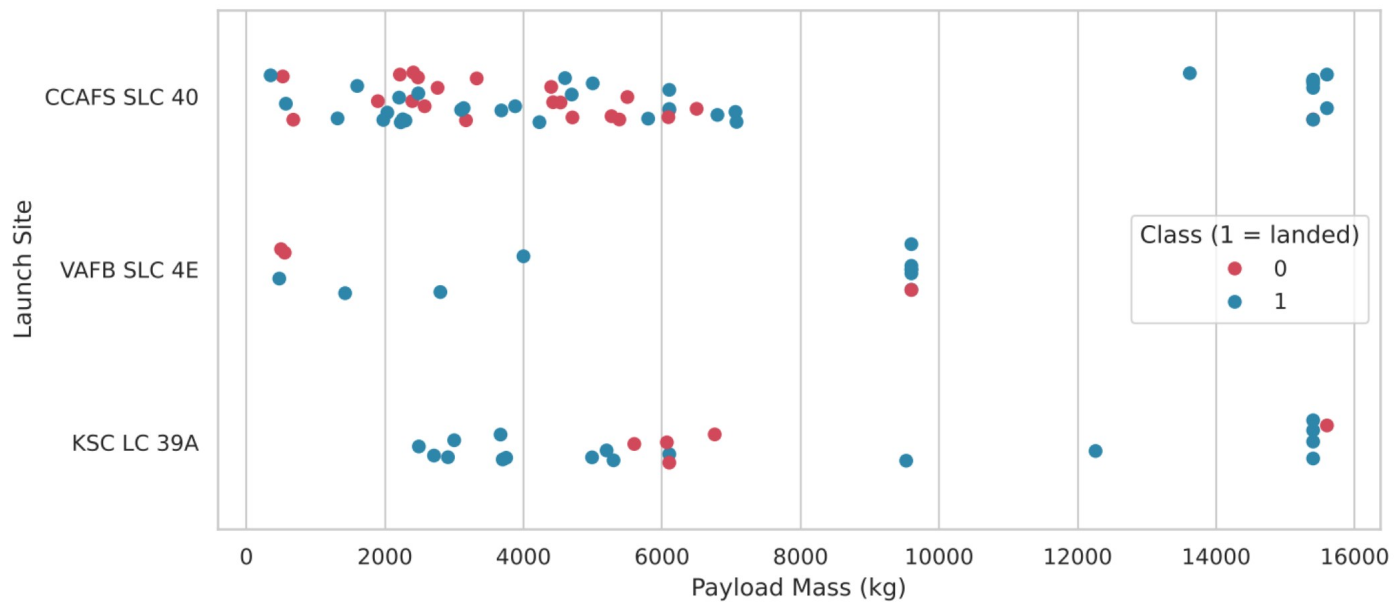
Red (failed) points cluster in the first ~20 flights, mostly at CCAFS SLC-40.

Every launch from flight 78 onward lands, so **experience matters more than site**.

CCAFS SLC-40 carries 55 of 90 launches.

EDA · PAYLOAD VS LAUNCH SITE

Heavy payloads above 10,000 kg rarely fail



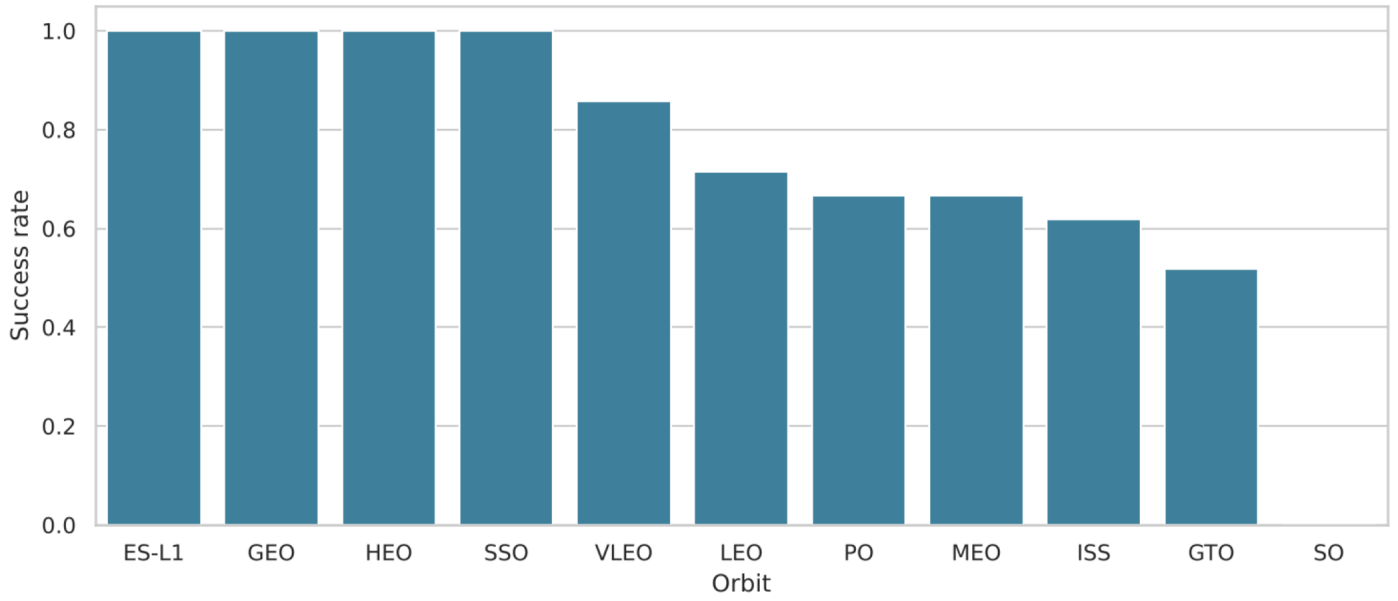
VAFB SLC-4E launches no payload above 10,000 kg.

The heaviest launches (13,600–15,600 kg, Starlink-class) come from the Florida sites and mostly land.

Mid-range payloads at CCAFS show the most mixed results.

EDA · SUCCESS RATE BY ORBIT

Four orbits land every time; GTO lands half the time



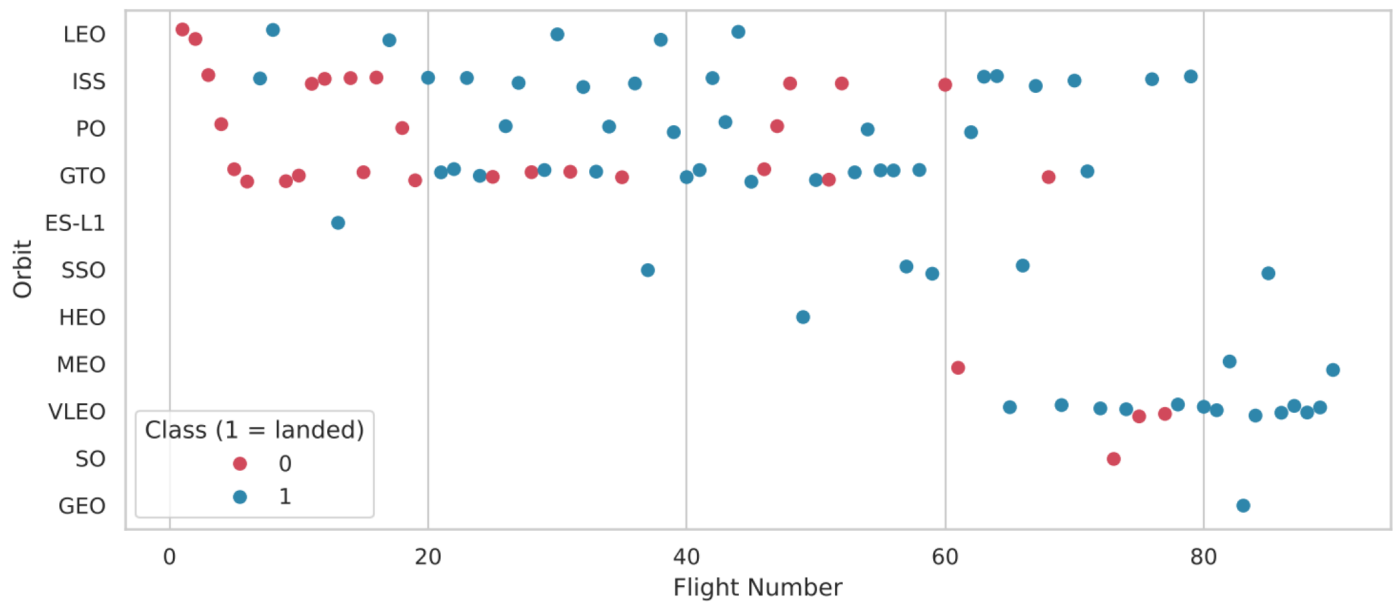
100%: ES-L1, GEO, HEO, SSO. Only SSO has more than one launch (5).

85.7%: VLEO (14 launches), the strongest high-volume orbit.

51.9%: GTO, the most common orbit and the hardest landing.

EDA · FLIGHT NUMBER VS ORBIT

LEO improves with experience; GTO does not



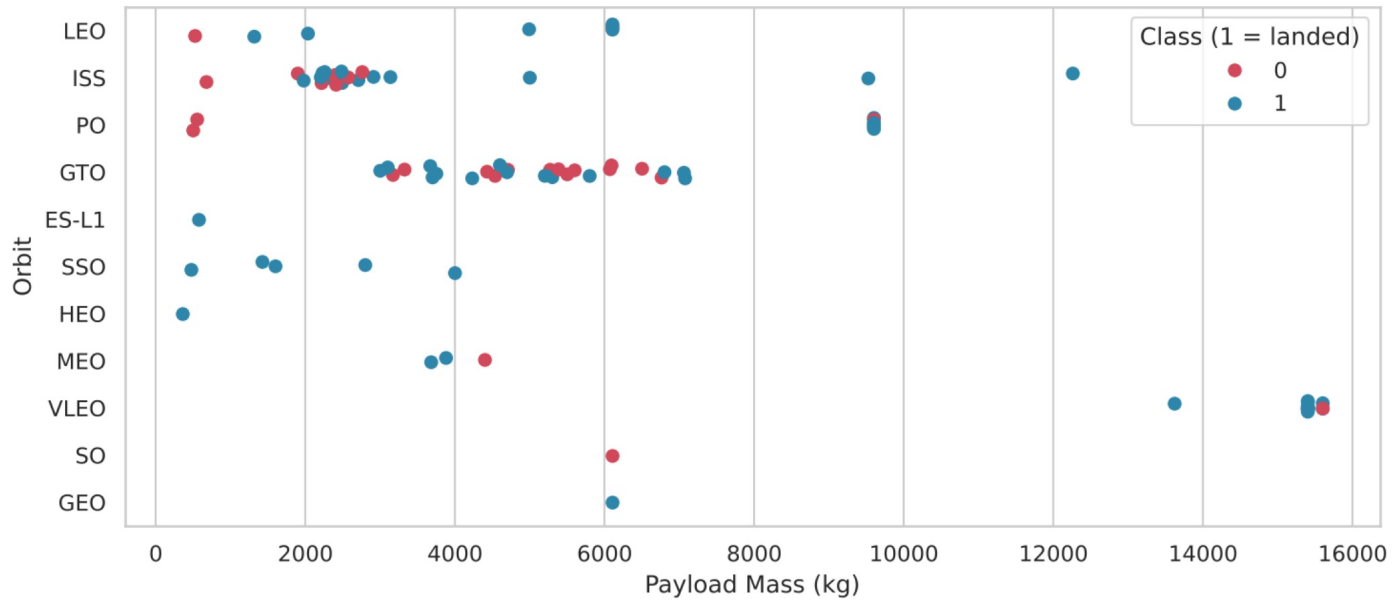
LEO failed only on flights 1 and 2, then landed every time.

GTO failures are spread across the whole flight range, so flight count does not explain them.

VLEO first appears at flight 65, always on Block 5 boosters.

EDA · PAYLOAD VS ORBIT

Heavy payloads land well for PO, ISS and VLEO



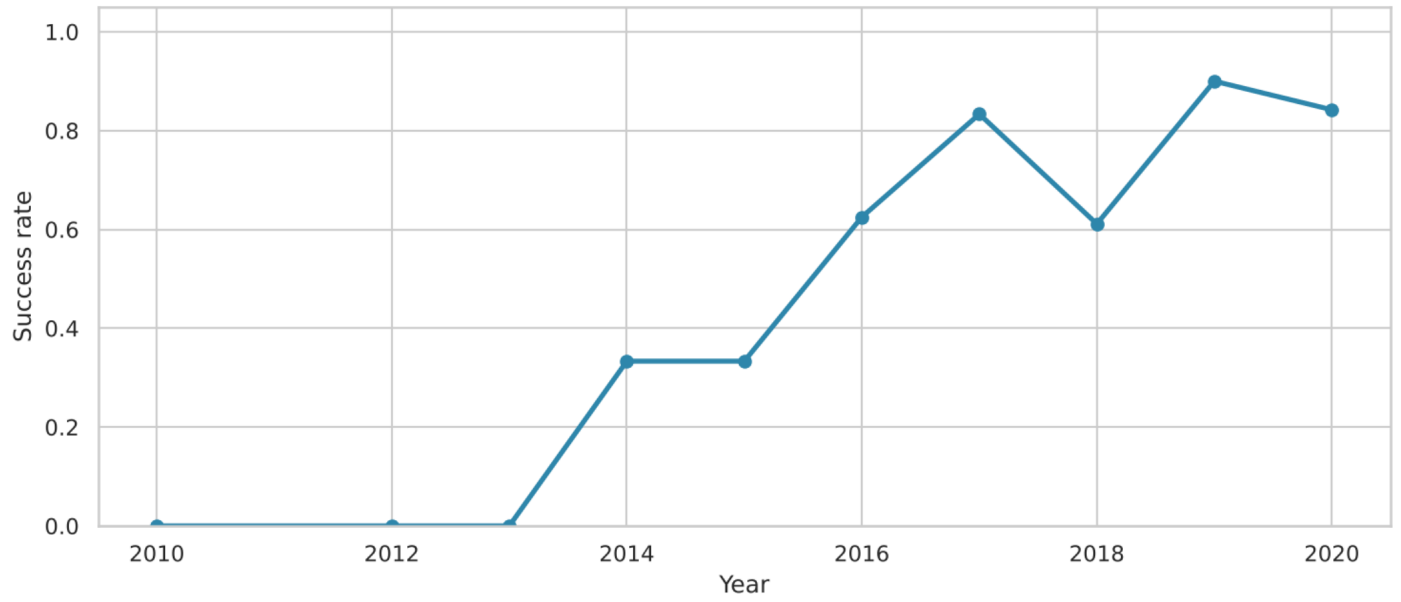
PO and ISS missions with heavier payloads mostly land.

GTO payloads sit in a narrow 3,000–7,000 kg band, with successes and failures mixed throughout.

VLEO carries the heaviest loads (13,600–15,600 kg) and still lands 12 of 14 times.

EDA · LAUNCH SUCCESS YEARLY TREND

Landing success climbed from 0% to 90% in six years



No successful landings from 2010 to 2013.

Steady gains from 2013 to 2017 (83%), a dip in 2018 (61%) when several reused boosters flew with no landing attempt, then **90% in 2019**.

2020 held at 84%.

EDA WITH SQL · LAUNCH SITES

Three launch sites, two of them in Florida

All launch site names

```
SELECT DISTINCT LaunchSite
FROM SPACEXTBL;
```

LaunchSite
CCAFS SLC 40
VAFB SLC 4E
KSC LC 39A

DISTINCT returns each site once.

Five records where site begins with “CCA”

```
SELECT * FROM SPACEXTBL
WHERE LaunchSite LIKE 'CCA%' LIMIT 5;
```

Date	Orbit	Payload kg	Outcome
2010-06-04	LEO	6,105	None None
2012-05-22	LEO	525	None None
2013-03-01	ISS	677	None None
2013-12-03	GTO	3,170	None None
2014-01-06	GTO	3,325	None None

EDA WITH SQL · PAYLOADS

CCAFS SLC-40 has carried 56% of all payload mass

Total payload mass by launch site

```
SELECT LaunchSite, SUM(PayloadMass)
FROM SPACEXTBL GROUP BY LaunchSite;
```

LaunchSite	Total payload (kg)
CCAFS SLC 40	305,151
KSC LC 39A	167,342
VAFB SLC 4E	76,953

Average payload, Block 1 boosters

```
SELECT AVG(PayloadMass)
FROM SPACEXTBL WHERE Block = 1;
```

2,578.8 kg

Early Block 1 boosters flew light payloads; the overall average is 6,105 kg.

EDA WITH SQL · LANDING MILESTONES

The first ground-pad landing came in December 2015

First successful ground-pad landing

```
SELECT MIN(Date) FROM SPACEXTBL
WHERE Outcome = 'True RTLS';
```

2015-12-22

Flight 17, booster B1019, LEO mission from CCAFS SLC-40.

Drone-ship successes, 4,000–6,000 kg

```
SELECT DISTINCT Serial FROM SPACEXTBL
WHERE Outcome = 'True ASDS'
AND PayloadMass > 4000
AND PayloadMass < 6000;
```

B1021	B1022	B1026	B1031	B1046	B1059

EDA WITH SQL · OUTCOMES AND BOOSTERS

Two of every three landings succeeded

Total successful and failed landings

```
SELECT Class, COUNT(*)
FROM SPACEXTBL GROUP BY Class;
```

60
Success (Class 1)

30
Failure (Class 0)

Boosters carrying the maximum payload

```
SELECT DISTINCT Serial FROM SPACEXTBL
WHERE PayloadMass = (SELECT
MAX(PayloadMass) FROM SPACEXTBL);
```

Booster serial	Payload (kg)
B1048	15,600
B1051	15,600

Both are Block 5 boosters on VLEO missions.

EDA WITH SQL · TIME ANALYSIS AND RANKING

Before 2017, most launches made no landing attempt

2015 drone-ship failures by month

```
SELECT substr(Date,6,2), Outcome,
Serial, LaunchSite FROM SPACEXTBL
WHERE Outcome = 'False ASDS'
AND substr(Date,1,4) = '2015';
```

Month	Outcome	Booster	Site
01	False ASDS	B1012	CCAFS SLC 40
04	False ASDS	B1015	CCAFS SLC 40

Outcomes ranked, 2010-06-04 to 2017-03-20

```
SELECT Outcome, COUNT(*) AS n ... WHERE
Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Outcome ORDER BY n DESC;
```

Rank	Outcome	Count
1	None None (no attempt)	9
2	True ASDS	5
3	False ASDS	4
4	True RTLS / True Ocean	3 each
5	None ASDS / False Ocean	2 each

03

Launch Sites Proximities Analysis and Dashboard

Folium maps, then the Plotly Dash app

FOLIUM · ALL LAUNCH SITES

Every launch site sits on a coastline

[Paste screenshot: Folium map with a circle and label at each launch site]

Site	Lat	Long
CCAFS SLC 40	28.562	−80.577
KSC LC 39A	28.608	−80.604
VAFB SLC 4E	34.632	−120.611

Florida sites launch east over the Atlantic and gain speed from Earth’s rotation; Vandenberg launches south over the Pacific for polar orbits.

FOLIUM · LAUNCH OUTCOMES BY SITE

Color-coded markers show where landings succeed

[Paste screenshot: MarkerCluster zoomed on a site, green = success, red = failure]

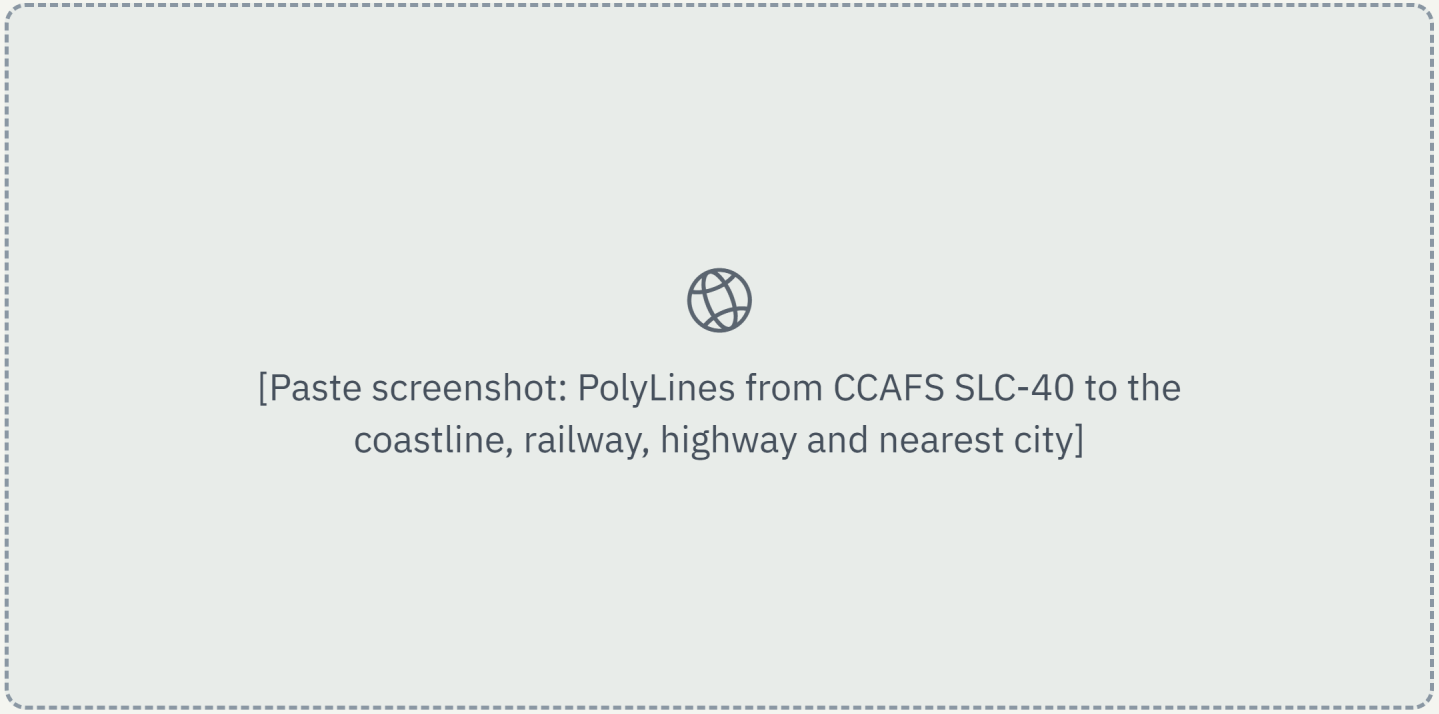
KSC LC 39A
17 of 22 · 77.3%

VAFB SLC 4E
10 of 13 · 76.9%

CCAFS SLC 40
33 of 55 · 60.0%

FOLIUM · PROXIMITY ANALYSIS

Close to the coast and transport, far from cities

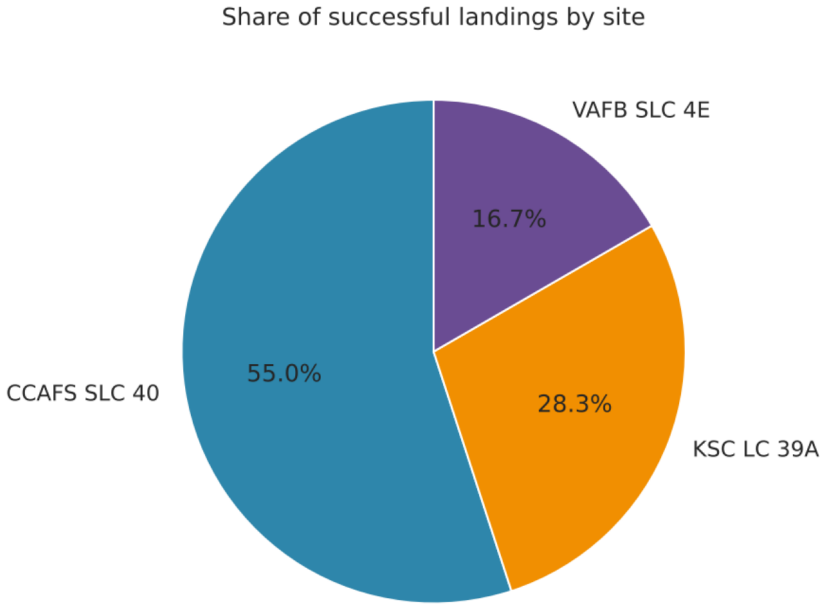


From CCAFS SLC-40 to	Distance
Coastline	0.93 km
Railway	1.33 km
Highway	0.65 km
Cape Canaveral (city)	17.62 km

Coast proximity lets spent stages and debris fall over water; rail and road move heavy hardware in; distance from cities keeps people safe.

PLOTLY DASH · SUCCESS BY SITE (ALL SITES)

CCAFS SLC-40 delivers the most successful landings



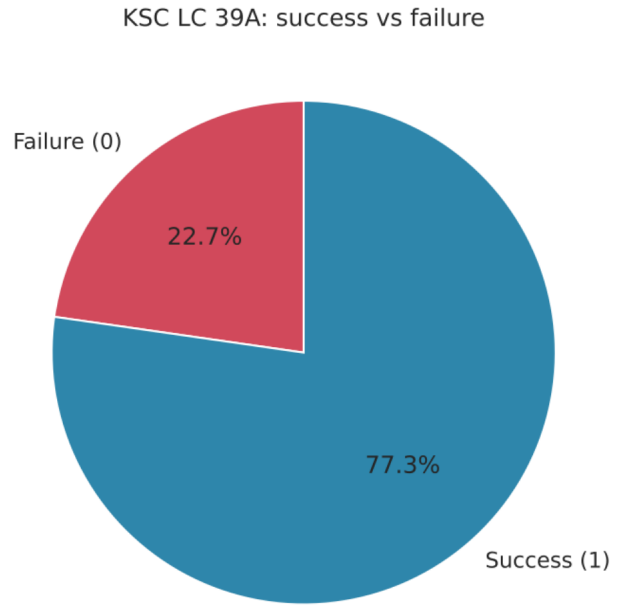
Of the 60 successful landings, **55% came from CCAFS SLC-40**, 28.3% from KSC LC-39A and 16.7% from VAFB SLC-4E.

This share reflects launch volume, not reliability: CCAFS flies the most launches but has the lowest success rate (60%).

Dropdown = “All Sites” in the Dash app.

PLOTLY DASH · SITE WITH HIGHEST SUCCESS RATIO

KSC LC-39A lands 77.3% of its boosters



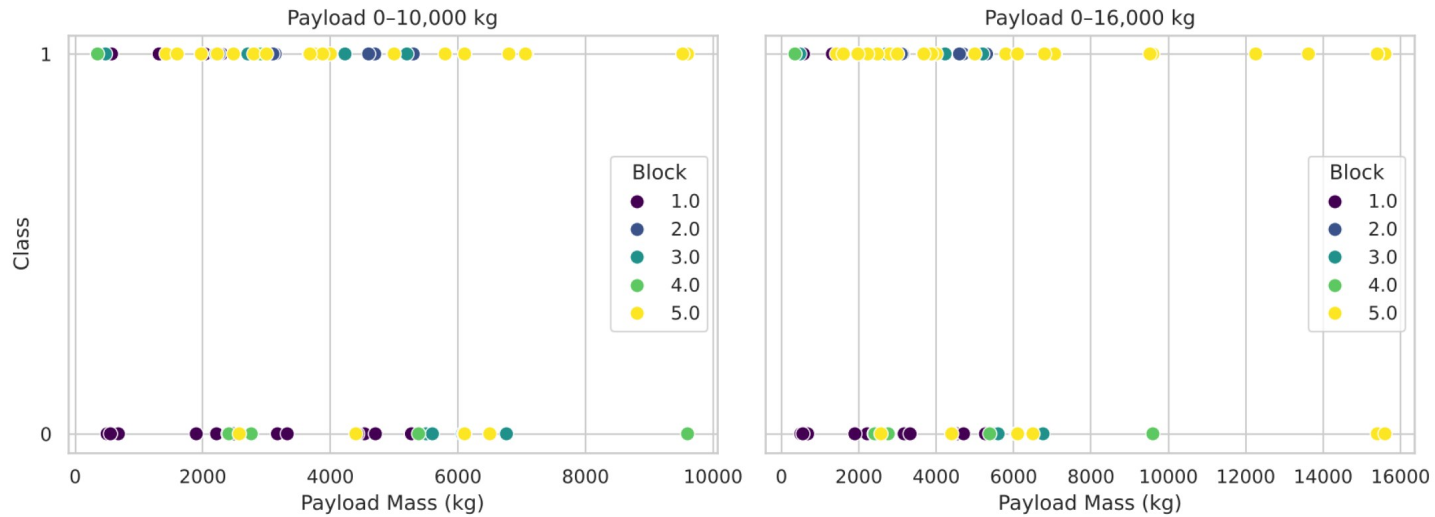
17 successes and 5 failures out of 22 launches.

VAFB SLC-4E is a close second at 76.9% (10 of 13).

Three of KSC’s five failures were GTO missions with no landing attempt.

PLOTLY DASH · PAYLOAD VS LAUNCH OUTCOME

Booster version matters more than payload mass



65.8% success under 4,000 kg

67.3% success at 4,000 kg and above

84.6% success on Block 5 boosters

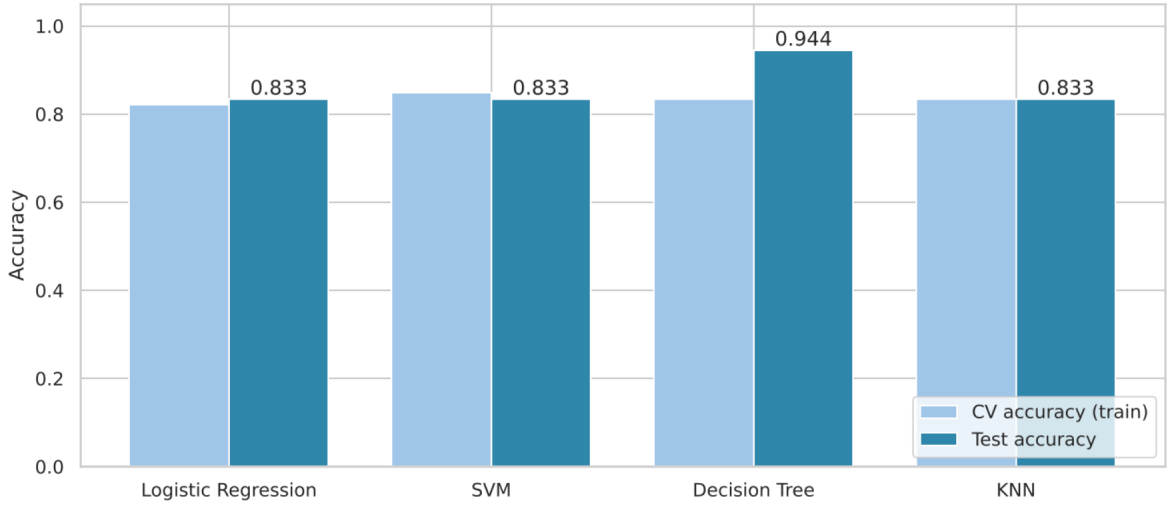
04

Predictive Analysis (Classification)

Four tuned models compared on held-out launches

CLASSIFICATION ACCURACY

The decision tree scores highest on the test set

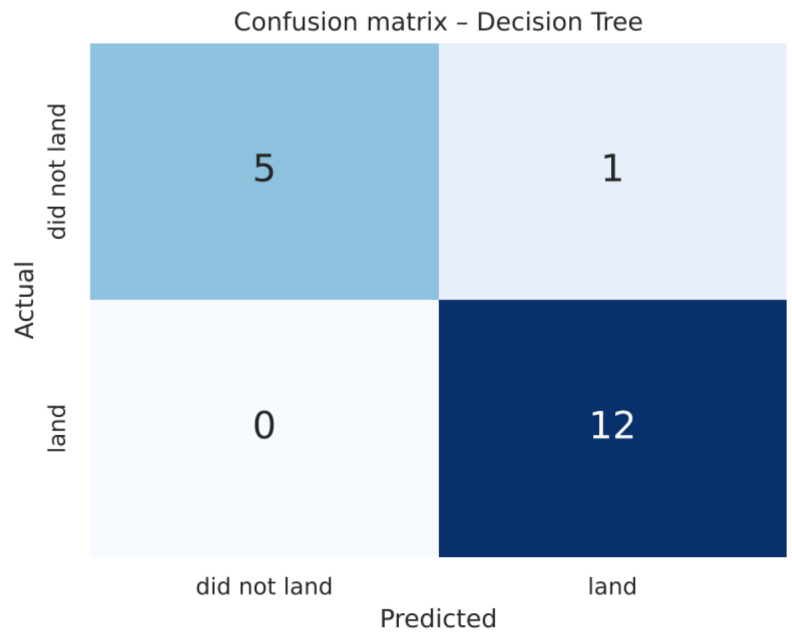


Model	CV acc.	Test acc.
Logistic Reg.	0.821	0.833
SVM	0.848	0.833
Decision Tree	0.834	0.944
KNN	0.834	0.833

Best tree: gini, max_depth 4, max_features sqrt, min_samples_leaf 4, min_samples_split 2. Test set = 18 launches, so one prediction moves accuracy by 5.6 points.

CONFUSION MATRIX · DECISION TREE

One mistake in 18 test launches



12

True positives: landings predicted correctly

5

True negatives: failures predicted correctly

1

False positive: a failure predicted to land

0

False negatives: no missed landings

The other three models each made 3 false positives. Every model’s weak spot is the same: telling failures apart from landings.

CONCLUSION

Landing outcome can be predicted well enough to price a launch

Experience drives success

Success grew from 0% before 2014 to 90% in 2019; later flight numbers land far more often.

Orbit matters

ES-L1, GEO, HEO and SSO landed every time; GTO, the most common orbit, only 51.9%.

KSC LC-39A is the most reliable site

77.3% success; CCAFS SLC-40 flies the most but lands 60%.

Decision Tree is the best model

94.4% test accuracy with no missed landings; the other three models tie at 83.3%.

CREATIVE INSIGHTS

What Space Y should take from this

63%

“Failures” are often choices

19 of the 30 failures were “None None”: no landing attempt. A model could separate intended expendable launches from real failures.

84.6%

Hardware generation is a lever

Block 5 boosters land 84.6% of the time versus 66.7% overall. Bidding should weight booster block heavily.

18

Small test sets need care

With 18 test launches, cross-validation scores (82–85%) are a fairer guide; retrain as new launches arrive.

APPENDIX

Data, code and notebooks

Dataset

dataset_part_2.csv: 90 Falcon 9 launches, June 2010 to November 2020, 18 columns including FlightNumber, PayloadMass, Orbit, LaunchSite, Outcome, Block, ReusedCount, Serial and Class.

Tools

Python, Pandas, NumPy, Requests, BeautifulSoup, Matplotlib, Seaborn, SQLite, Folium, Plotly Dash, scikit-learn.

Repository files

`jupyter-labs-spacex-data-collection-api.ipynb`
`jupyter-labs-webscraping.ipynb`
`labs-jupyter-spacex-Data-wrangling.ipynb`
`edadataviz.ipynb`
`jupyter-labs-eda-sql-coursera_sqllite.ipynb`
`lab_jupyter_launch_site_location.ipynb`
`spacex_dash_app.py`
`SpaceX_Machine_Learning_Prediction_Part_5.ipynb`

GitHub repository: github.com/gcastor/spacex-falcon9-capstone